

Theoretical and experimental research on the double lane change maneuver

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Abstract. A virtual double lane change test procedure is proposed based on the NATO AVTP 03-160W requirement. This test procedure simulates an emergency maneuver that determines the vehicle's dynamic stability and decisively influences road safety, being one of the most applied methods of performance evaluation. Using Trucksim modeling and simulation software, the virtual vehicle model and test procedure were established. The experimental data obtained with FED Alpha vehicle helped to validate both models. For this purpose, the vehicle dynamic responses such as steering wheel angle, vehicle lateral acceleration, yaw velocity and roll angle were comparatively analyzed. Important and valid conclusions concerning the optimization of the vehicle's dynamic behavior were drawn. Ultimately, a methodology to predict the maximum speed to successfully pass double lane change is proposed.

1 Introduction

One of the most significant causes for road accidents in Europe is related to vehicles stability, therefore, as mentioned in the European regional status report on road safety 2019 [1], one of the eight priority United Nations vehicle-safety standards refers to electronic stability program (ESP), which:

- prevents skidding and loss of control in cases of oversteering or understeering;
- is effective in reducing/avoiding single-car and roll-over crashes, reducing both fatal and serious injuries and saving lives.

Evaluating the performances of the complex system driver-vehicle-running surface has been applied on a large scale in automotive design, manufacturing and testing. Double lane change (DLC) is one of the most common methods used in this type of testing and evaluation, representing a simulation of an emergency maneuver performed to avoid an imminent impact, showing though the dynamic stability of a vehicle.

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The main procedures that describe the DLC maneuver, as detailed in [2], are the followings:

- ISO 3888 Part I and Part II - applicable for passengers' cars;
- Consumers Union Short Course (CUSC), based on tests developed by General Motors in 1960 years, being used for evaluating the ability to control a vehicle when sudden steering is needed;
- North Atlantic Treaty Organization (NATO) - Allied Vehicle Testing Publications - AVTP 03-160W, which is similar to ISO 3888 Part I and Part II, but applicable in general for military or militarized vehicles;
- Test Operations Procedure (TOP 2-2-609), which is similar to CUSC tests.

Specific technical data described in the above-mentioned testing procedures, related to the shaping and dimensioning of the track for DLC maneuver, are comparatively presented in fig. 1.

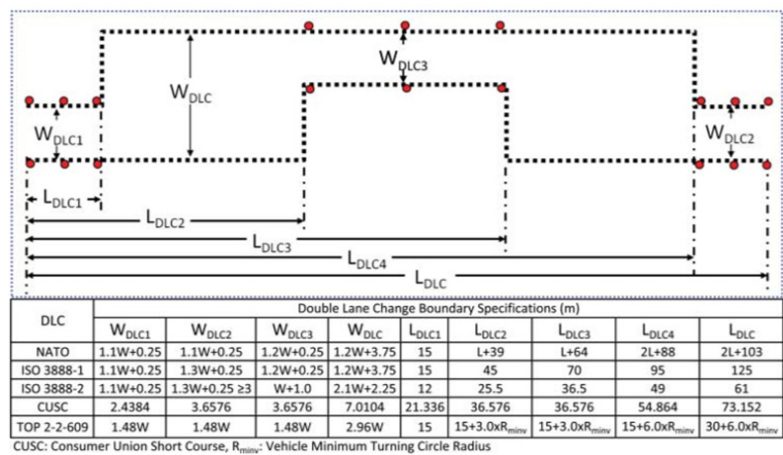


Fig. 1. Comparative analysis of the 5 DLC boundary specifications [2].

It can be noticed that the dimensions provided by CUSC test procedure are not dependent of vehicle dimensions, the ones from ISO 3888-1 and ISO 3888-2 are considering the vehicle width, while AVTP 03-160W and TOP 2-2-69 are function of both width and length of the vehicle under test.

Given the fact that the vehicle available for experiments is a military vehicle, the reference for the analysis presented in this paper is AVTP 03-160W, which consists in the following steps/conditions:

- 1) The vehicle will enter the track marked by cones with the lowest speed specified in the testing plan, moving straight forward through the 1st section (LDLC1 in fig. 1);
- 2) The vehicle will follow the path of the track by maintaining as much as possible a constant speed during the test;
- 3) The testing shall be repeated at different speeds specified in the testing plan, until the highest speed is achieved, until the stability limit of the vehicle is reached or until the path following without touching/hitting a cone becomes impossible;
- 4) Parameters of interest are recorded and the behavior of the vehicle is monitored during testing.

The literature analysis [3-5] shows a high interest on studying the stability and maneuverability of the vehicles, but there are not many studies concerning the dynamic behavior of a vehicle on both pavement and gravel tracks. The work presented in this paper aims to offer more insights on how a 4x4 combat vehicle behaves when performing the DLC maneuver on two different road surfaces and what the related implications are.

Therefore, as will be further detailed, chapter 2 outlines the main theoretical aspects regarding vehicle lateral stability considering rollover and skidding situations. Chapter 3 illustrates the field test procedure for both pavement and gravel test courses. The objective of chapter 4 is the development of a computer simulation test procedure and its validation by a comparison of results obtained by field tests and simulations with the real and the three-dimensional virtual 4x4 FED-Alpha (Fuel Efficiency Demonstrator Alpha) vehicle. Chapter 5 proposes a virtual procedure regarding the prediction of the maximum speed to successfully pass double lane change test. The last chapter emphasizes the conclusions and future approaches.

2 Theoretical considerations regarding vehicle lateral stability

A vehicle drivability defines its property to maintain the direction of rectilinear movement if the driver does not command any turn and to easily follow a cornering trajectory when the driver commands it. A more detailed definition of this notion, and not only, is presented in [6], where the differences in meaning between vehicle maneuverability and drivability are also clarified. On the other hand, stability is the property of vehicles to be less susceptible to skidding and rollover while driving and at rest. The National Highway Traffic Safety Administration, in its New Car Assessment Program (NCAP) [7], defines the rollover probability of a vehicle and establishes a rollover resistance rating system taking into consideration the static stability factor (SSF), defined as:

$$SSF = \frac{l_w}{2h} \quad (1)$$

where l_w is the track width of the vehicle and h is the height of the center of gravity of the vehicle, measured from the ground.

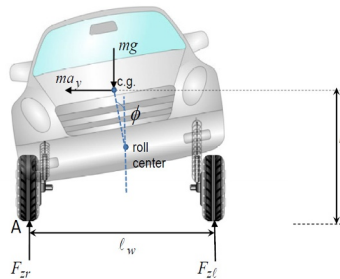


Fig. 2. Rollover moment balance in cornering [8].

According to fig. 2, if the moments about the right wheel contact point A are taken, the acceleration at which the left wheels lift off and the normal force F_{zl} becomes zero, has the following expression:

$$a_{y_lift-off} = \frac{l_w}{2h} g \quad (2)$$

Static stability factor has values ranging from 1.05 to 1.3, or more, and according to equation (2), the lower it is, the lower the lateral acceleration at which rollover occurs will be. The values of static stability factor and lift-off lateral acceleration for 4x4 FED-Alpha vehicle, taking into consideration its geometric dimensions, are as follows:

$$SSF = 1.22, a_{y_lift-off} = 1.22g \quad (3)$$

According to [9], for a vehicle designed with a SSF value of 1.2, or more, the critical roll angle is significantly greater than 10 degrees, when driving on a surface with a coefficient of friction of approximately 0.85. A more elaborate approach of this parameter is also presented in [9], where a dynamic analysis is provided by defining a new parameter, the dynamic stability factor, which is similar to SSF , but calculated at the point of vehicle rolling over.

It is also very important to mention the fact that before the occurrence of rollover phenomena, the road safety conditions require the appearance of vehicle skidding. Therefore, according to [10], the equation of vehicle speed calculated at the limit of skidding, assuming the fact that vehicle movement and road curvature are constant, is:

$$v_{skidding} = \sqrt{gR \frac{\varphi + \tan \beta}{1 - \varphi \tan \beta}}, \quad (4)$$

$$v_{skidding} = \sqrt{\varphi g R} \quad (\text{for a horizontal path, } \beta = 0) \quad (5)$$

where R stands for the instantaneous turning radius of the rear axle center, φ is the adherence coefficient of the road and β is the transverse elevation of the road.

Also, the equation of rollover speed as a function of static stability factor SSF , for a horizontal path, is described as following:

$$v_{rollover} = \sqrt{gR \frac{l_w}{2h}} \quad (6)$$

As emphasized before, by applying the condition of skidding before the occurrence of rollover, the following condition is mandatory:<

$$V_{skidding} < v_{rollover}, \text{ which implies } \varphi < SSF \quad (7)$$

which is an accomplished condition by all vehicles with a normal geometrical design.

Considering all the above aspects, the theoretical notions presented are simple ones, based on the simplified mathematical model of the vehicle, without considering the reducing effects of suspension or other systems such as ESP. Therefore, complementary dynamic maneuver tests are needed in order to determine the overall rollover resistance of a vehicle.

3 Field test procedure

The double lane change maneuver was performed using AVTP 03-160W as a reference, each standard point being tapped and accomplished. A high level of information was provided due to the AVT-248 Research Task Group, which activated under Applied Vehicle Technology, NATO Science & Technology Organization. The theoretical results were transposed in a Cooperative Demonstration of Technology (CDT) with the purpose to showcase the enhanced mobility prediction capabilities of Next-Generation NATO Reference Mobility Model. The activity took place at the Michigan Technical University / Keweenaw Research Center (MTU/KRC) in Houghton, the United States, in September 2018 [11].

Similar to AVTP 03-160W, test instrumentation data was provided first. The vehicle speed, the time and the distance were logged using VBOX 3i Dual Antenna RTK from Racelogic, while the roll angle and lateral acceleration were recorded from the BEI MotionPak Multi-Axis Sensor, model 160260-138 (fig. 3a). The turning angle of steering wheel was measured with a rotary sensor from Reinshaw Company. The installation of ring magnetic and read head is presented in fig. 3b, assuring a 408.6 pulses/deg effective angular resolution.



Fig. 3. a) Inertial Multi-Axis Sensor; b) Rotary Sensor.

Regarding the test courses, two tracks were made for the CDT at MTU/KRC, one on pavement (fig. 4a) and other on gravel (fig. 4b), with the coefficients of friction of 0.94 and 0.37, which were experimentally determined by the Saab Friction Test Vehicle. The cones were used to mark the lanes, their lay-out being given in fig. 1, with the track dimensions calculated based on vehicle length and width.



Fig. 4. a) Pavement track; b) Gravel track.

The tests were conducted based on the procedures described in AVTP 03-160W, the relevant steps and conditions being mentioned in this paper on chapter 1. Following this recommendation, speed, steering wheel angle, lateral acceleration and roll angle data was acquired as the FED-Alpha was driven over each course type at various speeds. A speed increment was used to reach the maximum speed laid down in the test plan. However, in order to protect the vehicle against damage, the test was not performed at the possible maximum speed.

4 Virtual vehicle model and test procedure validation

The virtual vehicle model and test procedure were established using Truksim modeling and simulation software [12]. The software delivers a set of tools to quickly simulate

vehicle dynamics in complex driving environments. The math model is parametric, and it has built-in equations that perform calculations using parameters and tables as placeholders in those equations. Values are provided to the vehicle simulation solver at runtime for hundreds of parameters and tables of numbers are delivered to define potentially nonlinear relationships between variables in the math model. The data provided at runtime causes the math model to simulate a specific vehicle under specific test conditions. Many of the state variables are defined within the math model with ordinary differential equations (ODEs) and equations of motion are used to calculate these derivatives.

With the modular configuration, it is possible to modify the parameters and run the simulation at any time, being very helpful to see and analyze how the change of one parameter influences the operation of the entire system. The specialized software allows the definition of the vehicle characteristics in detail, choosing and writing the values of the specific parameters for each sub-assembly. Consequently, the advantages that Trucksim software offers have led to its use in several scientific works, among which [13-15] can be mentioned.

The three-dimensional virtual vehicle model and the virtual environment can be observed in fig. 5a. The actual vehicle path was exported from the recorded GPS data during the experimental tests. VBOX File Processor software was used to compute X-Y coordinates from latitude and longitude using the first point as reference. Thus, the real trajectory (fig. 5b) has been set as input data, more specifically, as reference path. The math model used by the program includes a closed-loop steering controller that works by trying to follow the lateral coordinate defined as a function of distance along the path.

The initial speed was set at 72 km/h, similarly to real test, and a constant throttle control was applied. The virtual test procedure takes into consideration the real test conditions regarding the road surface. In this way, the measured value for the pavement friction coefficient is introduced in the program.

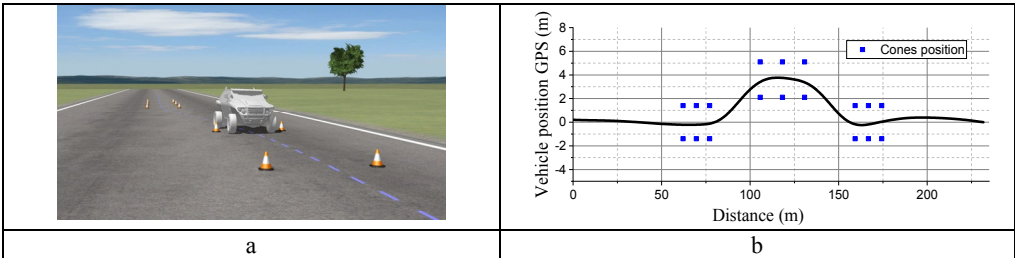
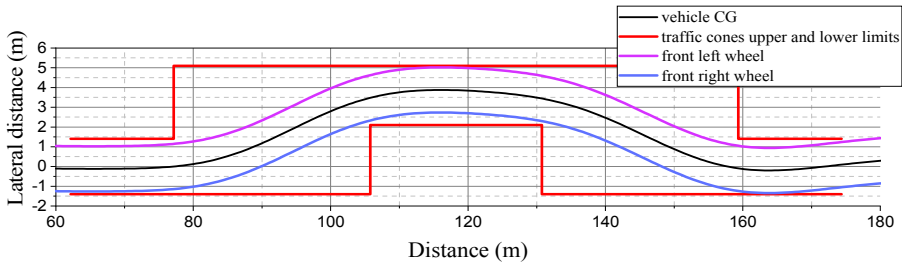


Fig. 5. a) Vehicle model in the virtual environment; b)Reference vehicle path.

Fig. 6 shows an overview of the accurate vehicle trajectory along the course, by overlaying the path boundaries according to cones position with the vehicle wheels path.



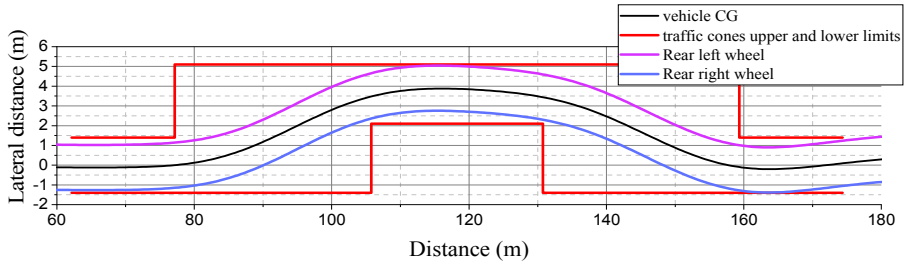


Fig. 6. Front wheels path and rear wheels path.

The validation of the virtual model was a very important process, especially because it was intended to be further used as an alternative in research and development. The method used considers comparison of results generated by the virtual model with those obtained during the experimental tests. The compared results are presented in fig. 7. It is important to emphasize the fact that both measured and simulated values of the parameters show a normal lateral dynamic response.

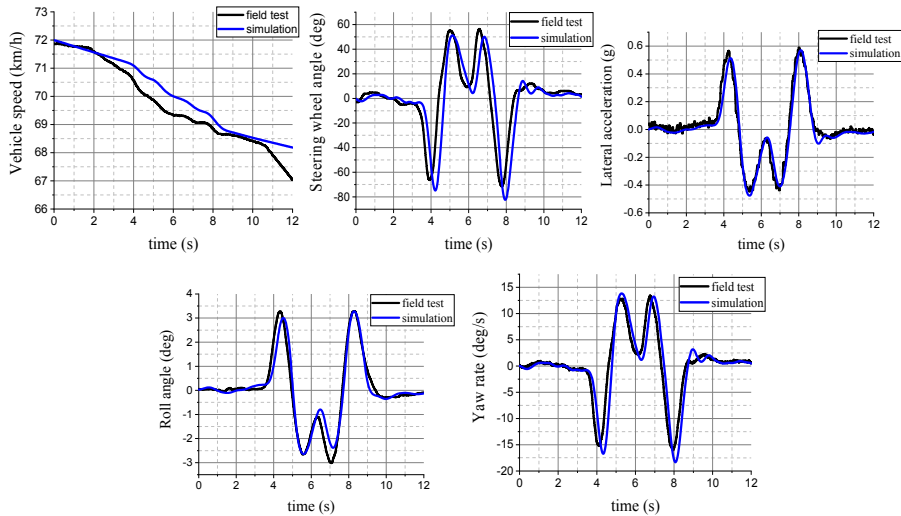
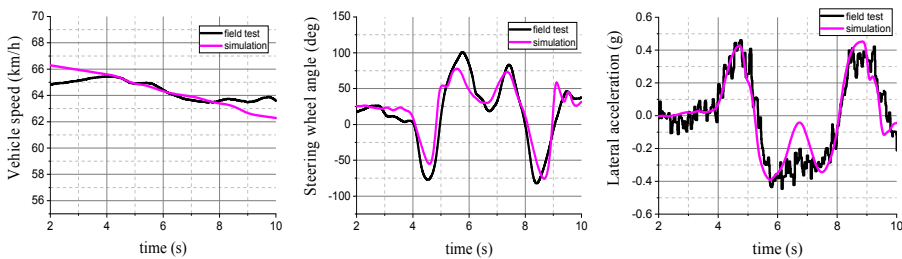


Fig. 7. Field test and simulation compared results – pavement, 70 km/h.

Similarly, a simulation of the double lane change maneuver was performed at the target speed of 65 km/h, for gravel and the compared results are presented in fig. 8. As it can be seen according to the higher values, the measured values of the steering wheel angle denote the possibly occurrence of skidding, due to a lower road surface adherence. It can be also noticed that the lateral acceleration values corresponding to the moment of direction change are lower than those on pavement, due to the lower speed.



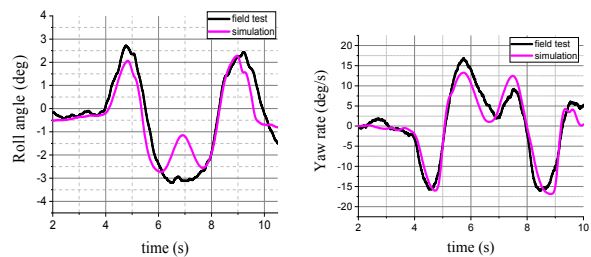


Fig. 8. Field test and simulation compared results – gravel, 65 km/h.

For a more precise interpretation of the compared results, the root mean square (RMS) and relative difference values are determined for both cases and presented in table 1 and table 2.

Table 1. RMS and relative difference values – pavement, 70 km/h.

RMS	Experimental data	Simulation data	Relative difference
Vehicle speed	69.742 km/h	69.835 km/h	0.13%
Lateral acceleration	0.229 g	0.212 g	-7.55%
Steering wheel angle	26.810°	27.006°	0.72%
Roll angle	1.458°	1.292°	-12.89%
Yaw rate	6.157°/s	6.462°/s	4.72%

Table 2. RMS and relative difference values – gravel, 65 km/h.

RMS	Experimental data	Simulation data	Relative difference
Vehicle speed	63.636 km/h	64.400 km/h	1.18%
Lateral acceleration	0.247 g	0.235 g	-5.39%
Steering wheel angle	49.279°	43.185°	-14.10%
Roll angle	1.868°	1.494°	-24.99%
Yaw rate	8.684°/s	8.270°/s	-5.04%

As it can be seen, there are some differences that encountered and could be caused by different error sources. The most significant of them are related as follows:

- the virtual model could not perfectly and precisely embrace the real vehicle complexity and its actual behavior;
- there should be considered the differences regarding the target speed which was in a much better manner maintained in the simulation than in the real test;
- the measurement errors of the testing devices must be also considered, especially the Global Positioning System (GPS) errors which directly influenced the reference path introduced as input data.

Both data sets presented above correspond to the maximum speed set in the testing plan. Taking into consideration the approximate turning radius corresponding to the reference vehicle path, theoretical values of the critical speeds are calculated based on (5) and (6) equations. Therefore, the vehicle speed calculated at the limit of skidding is approximately 100 km/h when driving on pavement and 70 km/h when driving on gravel. Also, the theoretical value of rollover speed is approximately 115 km/h. Analyzing these values, it can be noticed the great influence that the adherence coefficient of the road has on the skidding vehicle speed. However, for a better interpretation, the moment of skidding occurrence and the critical values of vehicle slip angle must be determined.

Using Trucksim software, the speed at which skidding, or rollover occurs can be determined, without taking into consideration the boundaries imposed by the DLC cones.

Instead, the vehicle side slip angle is considered. For example, for pavement, a maximum vehicle side slip angle of 2° is achieved at a speed of approximately 80 km/h, or 3.5° for 90 km/h. Also, the rollover becomes inherent after reaching a speed of 105 km/h. As regards gravel case, skidding starts at a speed of 65 km/h, while the vehicle becomes uncontrollable after reaching a speed of 80 km/h.

To sum up, the values obtained through simulations are much more realistic compared with the theoretical critical ones. Therefore, the advantages that the virtual vehicle model and test procedure offer are undeniable, especially when it comes to road safety. Moreover, the double lane change maneuver offers a much more precise method of vehicle dynamic behavior characterization through the boundaries imposed by the cones.

5 Determination of vehicle maximum speed to pass double lane change maneuver using Truksim

One of the advantages that a virtual model has is its use in addressing different aspects that were not completely/in more detail achieved through testing. The issue of predicting the maximum speed of approaching the double lane change maneuver has been addressed in various scientific papers [4, 16] and it is still of interest. The detailed results that can be obtained through the multitude of simulations lead to a much more accurate understanding of the dynamic behavior of the vehicle during the double change test. Thus, simulation data helps to easily identify the deficiencies encountered by the driver in approaching the test and to identify different solutions for optimizing the vehicle behavior.

Using Truksim software, it has been developed a new maximum speed prediction procedure based on the path constraints according to NATO standard test. In order to achieve this, four position sensors were set at the exterior lateral surface of tires and multiple conditions were established so that the cones not to be touched during the test.

A theoretical path was also established taking into consideration an ideal equidistant position of the vehicle towards the cones. The reference points set out were connected in the program using spline interpolation. The simulation was run in the loop using a speed increment of 1 km/h until the constraint conditions were no longer met. The overlaid vehicle paths are showed in fig. 9, which also illustrates in what manner the vehicle followed the target path. It is important to mention the fact that the driver preview time used for the simulation was 0.6 s, which is a compromise between following a path closely and also remaining stable. A shorter time will cause more aggressive steering, but a higher value will negatively influence sensitivity to abrupt changes.

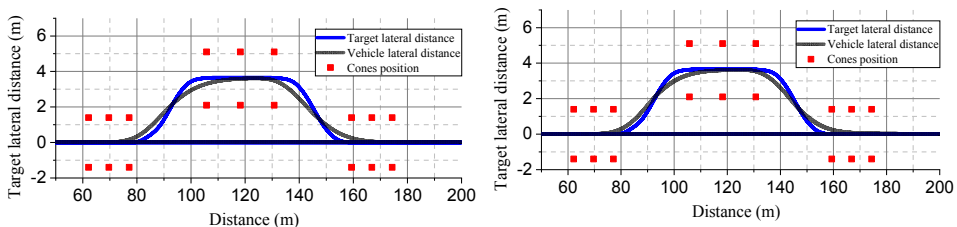


Fig. 9. Vehicle trajectory for pavement (left) and gravel (right).

Two simulations were conducted for both pavement and gravel tracks. The results obtained for the pavement track are illustrated in fig. 10. The maximum speed value at which the vehicle exceeded the limits imposed by the cones was 75 km/h. It can be observed that lateral acceleration and yaw rate values slightly increased, in accordance with the speed increment that was applied.

In a similar way, a simulation corresponding to gravel track was performed, starting from a lower value of the initial speed. The maximum speed at which the simulation stopped was found to be 67 km/h, a much lower value than the one found for pavement case study. An increase of vehicle dynamic responses on the second half of the course can be seen, due to the influence that the lower friction coefficient of the road had on vehicle's behavior. The results obtained for the gravel track are illustrated in fig. 11.

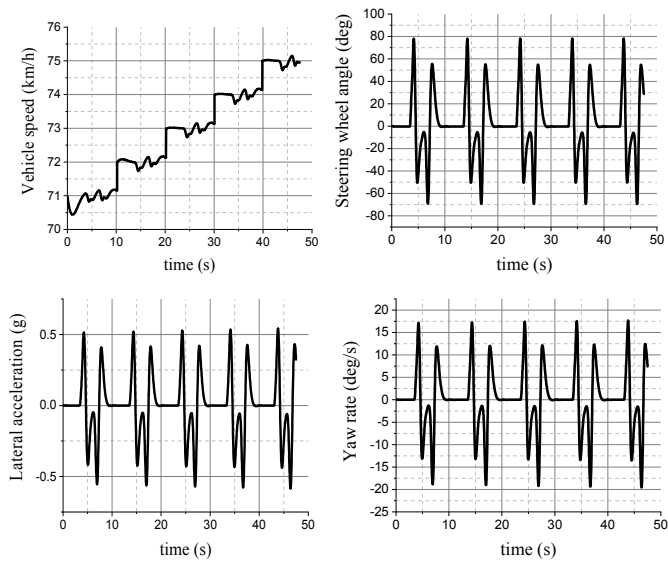


Fig. 10. Determination of maximum speed – pavement case study.

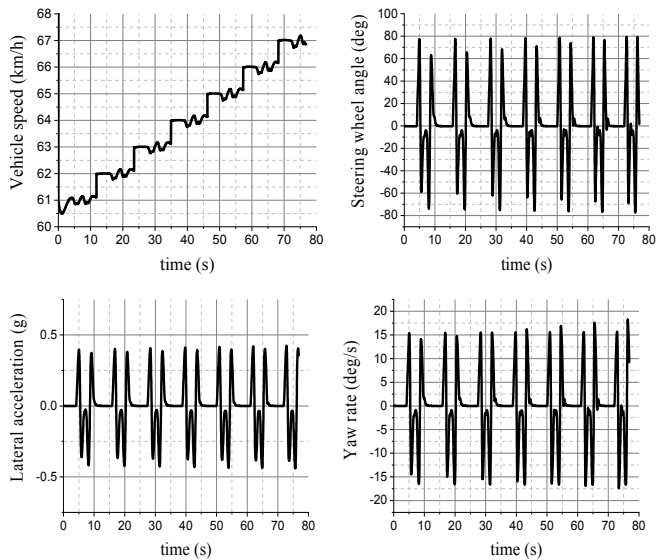


Fig. 11. Determination of maximum speed – gravel case study.

Analyzing all the above results, it can be emphasized the fact that the simulation results are comparable with the values measured during real testing. It is important to take into consideration that the increased values of the maximum speed reached in simulation are due to the chosen path and the accuracy of following it that the virtual model is offering. Within

the Cooperative Demonstration of Technology (CDT), virtual double lane change simulation tests were performed among others to provide the maximum speed to pass the test. The resulted values within CDT are similar to those obtained in the present paper, using Trucksim software, but a more detailed presentation can be found in [17].

In order to achieve higher values of vehicle speed to pass the double lane change maneuver, different aspects should be considered. As the speed increases, the lateral acceleration values will also increase with negative influences on the vehicle's dynamic behavior. Therefore, the optimization of the reference path which could provide an increase of speed has to be approached from the vehicle lateral dynamics perspective and considering the needs required by each specific application. For example, when it comes passengers comfort, beside the methods of reduction the rolling noise levels [18], lateral acceleration should be considerably limited, according to [19, 20].

6 Conclusions and future work

The double lane change maneuver is one of the most applied methods of performance evaluation when it comes to vehicle dynamics. This test procedure simulates an emergency maneuver that determines the dynamic stability of the vehicle which decisively influences road safety.

The novelty of the paper comes from the approaching of the double lane change maneuver considering two different road surface conditions and how they influence the vehicle dynamic responses. Moreover, one of the objectives of this paper was the development of a computer simulation test procedure and its validation by a comparison of results obtained by field tests and simulations with the real and the three-dimensional virtual 4x4 FED-Alpha vehicle. Skidding and rollover speeds were also determined through simulation, for both pavement and gravel cases, without considering the boundaries imposed by the cones. The values obtained were compared to the theoretical ones, being emphasized the importance of the virtual vehicle model and test procedure, especially when road safety aspects are approached.

The virtual model was further used to develop a virtual procedure with the aim of predicting the maximum speed to pass the double lane change test. The virtual procedure was performed for both pavement and gravel case studies and the obtained values were 75 km/s and 67 km/h, respectively. It can also be noticed that the maximum values obtained are lower than the theoretical critical ones, the advantage of the double lane change being also highlighted.

Further approaches concern path optimization methods with the purpose of increasing the maximum speed, but at the same time finding the best compromise between high dynamic performances and increased lateral stability. In order to develop a path optimization methodology, the proposed virtual procedure can be employed and improved by applying additional conditions regarding vehicle's dynamic responses such as lateral acceleration, roll angle or yaw rate. The control methods could have multiple applications, especially when it comes to autonomous vehicles or wheeled platforms for industrial applications.

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